

Calculus – Group Discussion

Week 2 – Differentiation Techniques

ADSAI & CS / PR12

Learning goals (Week 2)

- Compute derivatives using the product and quotient rules.
- Apply the chain rule to composite functions.
- Differentiate trigonometric functions and compositions.
- Perform implicit differentiation.
- Compute higher-order derivatives.

Basic rules of differentiation

Suppose two functions $f(x)$ and $g(x)$ are differentiable at $x \in \mathbb{R}$:

Power rule:

$$\text{If } f(x) = x^n, \quad \text{then } f'(x) = nx^{n-1}$$

Rule of linearity, for all scalars $\alpha, \beta \in \mathbb{R}$:

$$(\alpha f(x) + \beta g(x))' = \alpha f'(x) + \beta g'(x)$$

Product rule:

$$(f(x) g(x))' = f'(x)g(x) + f(x)g'(x)$$

Quotient rule:

$$\left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

Chain rule

Suppose $g(x)$ is differentiable at x and $f(u)$ is differentiable at $u = g(x)$.

If

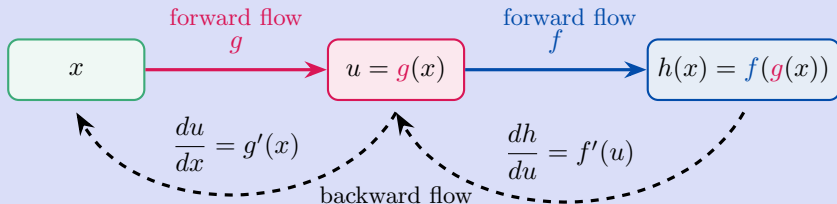
$$h(x) = f(g(x)),$$

then

$$h'(x) = f'(g(x))g'(x).$$

Equivalently,

$$\frac{d}{dx} f(g(x)) = \frac{df}{du} \frac{du}{dx} = f'(g(x)) \cdot g'(x).$$



Derivatives of special functions

Exponential functions:

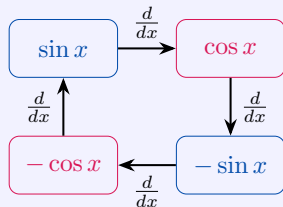
$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}a^x = a^x \ln(a)$$

Logarithmic functions:

$$\frac{d}{dx} \ln(x) = \frac{1}{x}, \quad \frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)}$$

Trigonometric functions:

$$\frac{d}{dx} \sin(x) = \cos(x), \quad \frac{d}{dx} \cos(x) = -\sin(x)$$



Higher-order derivatives

The derivative $f'(x)$ is itself a function, so we can (try to) differentiate it again.

Second derivative:

$$f''(x) = \frac{d}{dx} f'(x) = \frac{d}{dx} \frac{d}{dx} f(x) = \frac{d^2}{dx^2} f(x)$$

Third derivative:

$$f'''(x) = \frac{d}{dx} f''(x) = \frac{d}{dx} \frac{d}{dx} \frac{d}{dx} f(x) = \frac{d^3}{dx^3} f(x)$$

$\vdots$

n th derivative:

$$f^{(n)}(x) = \frac{d}{dx} f^{(n-1)}(x) = \underbrace{\frac{d}{dx} \frac{d}{dx} \cdots \frac{d}{dx}}_{n \text{ times}} f(x) = \frac{d^n}{dx^n} f(x)$$

Implicit differentiation

Sometimes a relation between x and y is not written explicitly as $y = f(x)$.

For example, $y^2 = x$ defines two differentiable functions $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$, with derivatives

$$\frac{dy_1}{dx} = \frac{1}{2\sqrt{x}}, \quad \frac{dy_2}{dx} = -\frac{1}{2\sqrt{x}} \quad (\text{Explicit derivatives})$$

To differentiate implicitly, treat y as a function of x by writing $y(x)$ and differentiate both sides of the equation using the chain rule:

$$\frac{d}{dx} [(y(x))^2] = \frac{d}{dx} x \quad \implies \quad 2y(x) \frac{d}{dx} y(x) = 1 \quad \implies \quad \frac{d}{dx} y(x) = \frac{1}{2y(x)}$$

Observe that this **implicit derivative** agrees with the derivatives of both **explicit solutions** $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$.

Key takeaways

- Start by identifying the structure of the function before choosing a differentiation rule.
- The chain rule is the key idea behind differentiating compositions, including trig, exponential, and logarithmic expressions.
- Higher-order derivatives can reveal useful patterns, such as the stability of e^x and the cyclic behaviour of $\sin(x)$.
- Implicit differentiation lets us work with equations where y is not isolated.
- Clean notation and factoring often make derivative formulas easier to interpret.